

Autonomous AI Support & Agent Copilot

Scaling Customer Operations for 50M+ Users via Real-Time API Tool-Calling & RAG

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[Portfolio](#) | [Email](#)**45%**

TICKET DEFLECTION

<2 MinFIRST RESPONSE TIME
(FROM >6H)**2.0 → 4.2**

CSAT LIFT (/5.0)

1-2 HrsRESOLUTION TAT (FROM 3-4
DAYS)

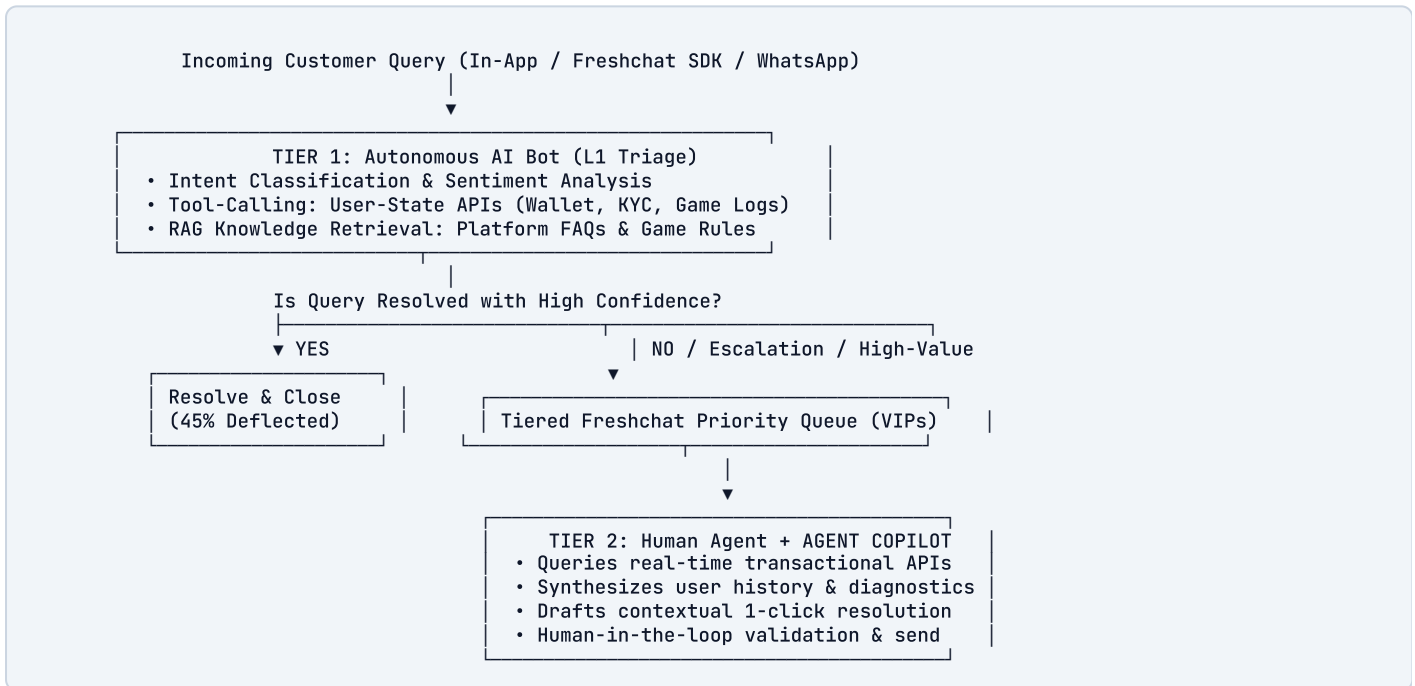
1. STRATEGIC CONTEXT & THE PROBLEM

At **Witzeal Technologies**, our consumer gaming platform scaled rapidly to **50M+ registered users and 300K+ DAU**. However, support operations were on the verge of breakdown: a small operations pod of 3 human agents was bombarded by thousands of daily tickets, resulting in a **First Response Time (FRT) exceeding 6 hours** and resolution turnarounds lagging between **3-4 days**.

- **Repetitive, Low-Intent Volume:** 40%-50% of incoming tickets were routine queries regarding balance reconciliation ("Where is my money?"), withdrawal statuses, or game rules.
- **Agent Cognitive Fatigue:** Human agents spent 70% of their shift manually pulling data from backend database consoles (checking KYC status, transaction logs, and wallet ledgers) before typing standard responses.
- **Direct Business Churn:** Data modeling revealed a direct causal correlation between support resolution latency (>12 hours) and paid high-depositor churn.

2. PRODUCT SOLUTION & TWO-TIER AI ARCHITECTURE

Rather than deploying a generic conversational chatbot that merely generates plausible answers, I designed an **agentic two-tier architecture** that combined deterministic backend API tool-calling with generative language models.



- **Tier 1 — Autonomous L1 Triage & Deflection Bot (Freshchat):** Integrated directly into Freshchat to intercept incoming chats. The agent identifies intent, invokes tool-calling against internal microservices to fetch live transactional states (e.g., "Withdrawal processed via IMPS Ref #9281, bank clearance pending"), and resolves queries in real time.
- **Tier 2 — AI Support Copilot ("Agent Helper") for Support Pod:** For complex, high-roller, or dispute escalations, the ticket is routed to human specialists. The Copilot automatically queries user-state APIs in the background and populates the agent console with synthesized diagnostics and a pre-drafted resolution response for one-click human review.

Autonomous AI Support & Agent Copilot (Cont.)

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3. TECHNICAL ENGINEERING, TOOL-CALLING & SAFETY GUARDRAILS

In financial transactions and gaming, accuracy is non-negotiable. Hallucinating a transaction confirmation or quoting incorrect ledger balances would create regulatory violations and player distrust.

- **Deterministic API Tool-Calling vs. Generative Output:** We strictly separated data retrieval from text synthesis. The LLM was prohibited from fabricating financial numbers. Tool-calling returned structured JSON payloads (e.g., `{status: "SUCCESS", amount: 500, utr: "90218"}`), and the LLM's only role was formatting that data into clear, empathetic user copy.
- **RAG Pipeline for Rules & Regulatory Policies:** Embedded game rules, TDS tax deduction guidelines (Section 194BA), and KYC compliance policies into a vector store. The bot performed semantic search to ground all policy answers in verified documentation.
- **Automated Escalation Ladders:** Designed confidence-score thresholds. If query intent confidence dropped below 85%, or if user sentiment exhibited severe frustration ("fraud", "cheat", legal threats), the system immediately bypassed the bot and triggered an urgent human handoff.
- **PII Masking & Toxic Content Filtration:** Implemented safety filters stripping PAN numbers, bank account details, and profanity before payload processing.

4. MEASURABLE BUSINESS OUTCOMES & ADOPTION

Core Metric Shift: In less than 60 days post-deployment, customer support transformed from a major operational bottleneck and churn driver into an automated, high-CSAT competitive advantage.

- **45% Autonomous Ticket Deflection:** Nearly half of all incoming support volume was completely resolved by the autonomous bot without human touch, freeing up agents for VIP retention and edge-case disputes.
- **Sub-2 Minute First Response Time (FRT):** Slashed initial customer wait times from **>6 hours to under 2 minutes**, creating immediate trust at the critical moment of user inquiry.
- **CSAT Increased from 2.0 to 4.2 / 5.0:** Player sentiment inverted as fast, accurate answers replaced days of waiting.
- **Resolution Turnaround Time (TAT) Cut to 1-2 Hours:** Standard ticket resolution collapsed from **3-4 days to 1-2 hours**, and high-priority VIP escalation queues were cleared within <12 hours.
- **Zero Headcount Expansion:** The 10+ member support team absorbed a 2.5x increase in platform DAU without hiring additional support personnel.

5. KEY PM LEARNINGS FOR BUILDING AI AT SCALE

- **"AI in operations must be tool-integrated, not just conversational."** A chat interface without access to internal system state is just an FAQ search engine. Connecting the model to live transactional APIs via tool-calling is what transforms AI into an autonomous operational workforce.
- **Human-in-the-Loop (HITL) accelerates internal adoption.** Support agents initially feared AI replacement. By framing the internal tool as a "Copilot" that drafts answers and removes database lookup drudgery, adoption hit 100% within two weeks.
- **Measure outcomes, not model benchmarks.** The success of the AI team wasn't measured by perplexity or BLEU score; it was measured by *First Response Time, Ticket Deflection Rate, Agent Handle Time, and CSAT*.